Building Pakistan's grid, line by line

Intro: Since 1998, the National Transmission & Despatch Company (NTDC) has carried electricity from power plants to the cities, farms and factories that need it. Today the company operates as the National Grid Company of Pakistan (NGC), the legal successor of NTDC. This page lists the projects we have delivered, the ones under construction, and the ones planned for the next three years.

Buttons: View completed projects | View current projects | View future plans

At a glance

- **1998:** NTDC incorporated on 6 November and began commercial operation on 24 December, taking over the 220 kV and 500 kV network from WAPDA.
- **18:** 500 kV grid stations operated by NGC.
- **PKR 750 billion:** board-approved investment programme through FY 2028.
- **878 km:** length of the Matiari–Lahore HVDC line, Pakistan's first.
- **765 kV:** the new voltage level now being introduced to the national grid.

Who we are now

In November 2024 the federal government restructured NTDC into three companies. The National Grid Company of Pakistan owns and maintains the transmission network. The Independent System and Market Operator runs the grid in real time and the electricity market. The Energy Infrastructure Development and Management Company focuses on developing new infrastructure and attracting private investment. Each has its own board.

Completed projects

Matiari–Lahore ±660 kV HVDC line

Status: Operating since 1 September 2021

An 878 km high-voltage direct current line from Matiari in Sindh to Nankana Sahib near Lahore. It is the country's first HVDC line and first private-sector transmission project, built under CPEC with a design capacity of 2,000 MW. It moves power from the south, where much of the generation is, to the load centres of the north.

Matiari 500 kV connections

Status: Completed

NTDC planned ten 500 kV circuits to connect southern power plants to the Matiari converter station. Eight had been energised by April 2021, including two circuits from the 1,320 MW Port Qasim coal plant.

Thar–Matiari 500 kV line

Status: Completed in 2023

A 220 km double-circuit line connecting the 1,320 MW coal plant in Thar to the grid at Matiari. NTDC built it in about two and a half months for roughly PKR 15 billion from its own resources.

Polan–Jiwani line

Status: Completed in 2023

A transmission line that allows Gwadar to import 100 MW of electricity from Iran.

Suki Kinari 500 kV line

Status: Completed

Carries power from the 884 MW Suki Kinari hydropower project to the national grid.

K2/K3–Matiari and KKI–Port Qasim 500 kV circuits

Status: Energised in 2025

Additional 500 kV circuits that strengthen the network around Karachi and Matiari.

500 kV Faisalabad West grid station

Status: Phase I energised

Built at a cost of PKR 9.379 billion with allied transmission lines. It relieves overloading and low voltage in the Faisalabad region.

Gharo–Jhimpir wind corridor grid station

Status: Completed

A 220/132 kV grid station with 35 km of lines, financed with PKR 6.4 billion, to bring 1,274 MW from ten wind projects into the grid. It is designed so it can later be upgraded to 500 kV.

Projects completed since November 2025

Status: Completed

NGC reported nine major completions in six months:

- 500 kV Lahore North grid station, which improves voltage in Lahore and lets the HVDC line be used to its full potential
- 220 kV Quaid-e-Azam Business Park grid station, which supplies a special economic zone in Punjab
- Interconnection of the Ghazi Barotha–Gatti line with the 500 kV Allama Iqbal Industrial City grid station in Faisalabad
- 500 kV Nowshera transmission line
- 55 km 220 kV Faisalabad West–Lalian line
- Six autotransformer upgrades (160 to 250 MVA) at Nokhar, Okara, Yousafwala and Hyderabad grid stations

Other recent upgrades

A fourth 250 MVA autotransformer at 500 kV Sheikhupura and a new autotransformer at 220 kV Daud Khel.

Current projects

Dasu transmission system, 765 kV

Status: Under construction

Pakistan's first 765 kV network, built to carry power from the 2,160 MW first stage of the Dasu hydropower project. It has four parts:

- **Dasu to Mansehra line:** 157 km, about 419 towers. Physical progress is 46%, with completion currently expected by 30 June 2028.
- **Mansehra 765/220 kV substation:** the country's first 765 kV substation, under construction by PowerChina since December 2023.
- **Mansehra to Islamabad West line:** 97.6 km.
- **Islamabad West 765/500/220/132 kV substation:** financed by the World Bank and NTDC equity. It will also carry power from Suki Kinari and the Tarbela 5th extension. The project's closing date is 31 December 2026.

CASA-1000, Pakistan section

Status: Under construction

Pakistan's part of the regional link that will bring up to 1,300 MW of hydropower from Central Asia. It includes the HVDC converter station at Nowshera and a 113 km HVDC line from Torkham to Nowshera. Delivery also depends on the Afghan section of the project.

National Transmission Modernization Project (NTMP-I)

Status: Closing, end of 2026

A World Bank-supported programme with three parts: upgrading and expanding the network (including the new Islamabad West and Nowshera 500 kV substations), a company-wide ERP system, and technical assistance and training.

SCADA-III

Status: Partly live

A new control room and SCADA system at the National Power Control Centre in Islamabad went partly live on 20 February 2025. The project also adds an Energy Management System and a Generation Management System, so faults are found and fixed faster. Live-line installation of optical ground wire (OPGW) fibre has reached 1,030 km of lines nationwide. In October 2026 NGC directed power plants to restore telemetry links to reach at least 95% data quality.

Future plans

Investment plan, FY 2026 to FY 2028

NGC's plan of about PKR 750 billion groups projects into five areas: special economic zones, constraint removal, system expansion, power evacuation and other development. It builds on the PKR 352 billion plan approved by the regulator, NEPRA.

New grid stations and lines

- 220 kV Dhabeji grid station for the special economic zone near Karachi
- 220 kV Haripur substation
- 220 kV Swabi grid station, fed from 500 kV Nowshera
- 500 kV Allama Iqbal Industrial City grid station, for a planned 600 MW of demand
- 220 kV Dera Ismail Khan–Zhob line, to cut system losses
- 220 kV Mirpur Khas grid station with allied lines
- Extra transformer capacity at existing 500 kV and 220 kV grid stations

Grid stability

Switched shunt capacitors and bus reactors at stations including Lahore South, Sheikh Muhammadi, Sarfraz Nagar, Loralai, Shikarpur and Zhob, planned for 2026-27, to keep voltage steady.

Power for the next big dams

The ten-year expansion plan (2024-34) includes evacuation for the Tarbela 5th extension (1,530 MW) and Diamer-Bhasha (4,500 MW), with a 765 kV line from Dasu to Lahore North planned for 2028-29.

Solar

NTDC has acquired 2,553 acres in Muzaffargarh for a 600 MW solar power project.

Why these projects matter

Power is generated far from where it is used: about three-quarters of demand is in the central and northern regions, while much new generation is in the south and the mountains. Our projects move power across that distance, reduce losses, and keep the grid stable as new hydropower, wind, coal and imports join it.

Related announcements

- Signing of the tripartite agreement for the US$ 2.4 billion, 1,124 MW Kohala hydropower project (CPEC)
- Agreement for the US$ 1.5 billion Azad Pattan hydropower project
- Board approval of an indigenization policy to encourage local industry
- New standard bidding documents for goods, consulting services, construction works and EPC contracts

Contact

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Notes for the web team

Statuses and dates come from news reports, regulator filings and project documents and may have changed. Confirm each project with NGC before publishing, especially Islamabad West (target dates have slipped before), the Dasu line, CASA-1000 and any item marked as planned. Add the official sources: ntdc.gov.pk, NEPRA filings, the World Bank NTMP-I documents and the CASA-1000 project site.